

AI-Assisted Linear Regression

1.1 Project Setup and Business Problem Understanding

- I'm going to work on an ML case study focused on EDA, data preprocessing, feature engineering, and building a Linear Regression model. That's the scope for now. Can you help me set up a minimal project directory structure?

Note: Place ElectraHub_Data.csv into data/raw/ before moving to the next step. Open Business_Problem.docx and read through the business context and objective before continuing.

1.2 Data Understanding

- Create a new notebook named 01_data_understanding.ipynb and place it under the notebooks folder.
- @01_data_understanding.ipynb I've placed the dataset in data/raw/. Can you load the dataset and add a structured summary to the notebook? Also, execute it and include details such as the number of rows and columns, what each column represents, the data types, the range of values for numerical columns, and any initial observations worth flagging before we start cleaning the data.
- What is the full price range of Product_Price in this dataset? Can you show the minimum, maximum, mean, and a count of products priced under \$50?

1.3 Data Preprocessing

- Create a new notebook named 02_preprocessing.ipynb and place it under the notebooks folder.
- @02_preprocessing.ipynb Can you check the data quality of the dataset in data/raw/ and add the findings to the notebook and also execute it? I want to know if there are any missing values, whether the data types look correct for each column, and whether there are any values that look suspicious or out of place given what each column is supposed to represent.
- @02_preprocessing.ipynb Based on what you found, can you suggest a cleaning plan and add it to the notebook? For each issue, tell me what you would recommend and why. Please do not run anything yet — I want to look at the plan before we go ahead.
- @02_preprocessing.ipynb The plan looks good. Can you apply the cleaning steps only — not any encoding or other methods — and add the code to the notebook? Save the cleaned dataset to data/processed/ and print a confirmation of the shape and column types after cleaning.

1.4 Univariate EDA

- Create a new notebook named 03_eda_univariate.ipynb and place it under the notebooks folder.
- @03_eda_univariate.ipynb Can you plot a histogram with a smooth curve on top for the Sales column and add it to the notebook? Include vertical lines showing the mean and median. I want to understand the shape of the distribution. Also, execute the notebook.
- @03_eda_univariate.ipynb Can you add a histogram and a box plot side by side for Advertising_Expenditure to the notebook? I want to see how the values are spread, whether there are any outliers, and how symmetric the distribution looks. Also, execute the notebook.
- @03_eda_univariate.ipynb Can you add a bar chart for the Popularity column to the notebook? Order the bars from Very Low to Very High — not alphabetically. Also, execute the notebook.

Note: The same approach can be applied to other variables: Product_Price, Campaign_Engagement_Score, Discount_Percentage, and Return_Rate.

1.5 Bivariate EDA

- Create a new notebook named 04_eda_bivariate.ipynb and place it under the notebooks folder.
- @04_eda_bivariate.ipynb Can you add a scatter plot of Advertising_Expenditure vs Sales to the notebook? Just the raw scatter, no lines. I want to see the general shape of the relationship between these two variables.
- @04_eda_bivariate.ipynb Can you add a scatter plot of Campaign_Engagement_Score vs Sales to the notebook? Again, just the raw scatter. I want to compare the pattern here to the one we saw with Advertising_Expenditure. Also, execute the notebook.
- @04_eda_bivariate.ipynb Can you add a scatter plot of Discount_Percentage vs Sales to the notebook? I want to see whether higher discounts tend to come with higher or lower sales. Also, execute the notebook.
- What is the actual range of Discount_Percentage in this dataset? Show the minimum, maximum, and mean. Is there any product with zero discount?
- @04_eda_bivariate.ipynb Can you add a box plot of Sales grouped by Popularity level to the notebook? Order the groups from Very Low to Very High on the x-axis. Also, execute the notebook.
- @04_eda_bivariate.ipynb Can you add a box plot of Sales grouped by Campaign_Type to the notebook? I want to see whether different campaign types are associated with different sales levels. Also, execute the notebook.

1.6 Multivariate EDA

- Create a new notebook named 05_eda_multivariate.ipynb and place it under the notebooks folder.
- @05_eda_multivariate.ipynb Can you add a correlation heatmap to the notebook for all numerical variables including Sales? Show the correlation values inside each cell so they are easy to read. Also, execute the notebook.
- @05_eda_multivariate.ipynb Can you add a horizontal bar chart to the notebook showing all numerical features ranked by their correlation with Sales, from strongest positive to strongest negative? This will help us decide which variables to model first. Also, execute the notebook.
- @05_eda_multivariate.ipynb Can you add a pairplot to the notebook for Campaign_Engagement_Score, Advertising_Expenditure, Avg_Customer_Rating, Discount_Percentage, and Sales? I want to visually scan the relationships between these variables at the same time. Also, execute the notebook.
- @05_eda_multivariate.ipynb Can you add a grouped bar chart to the notebook showing average Sales by Campaign_Type, with bars grouped by Region? I want to see if certain combinations of region and campaign type stand out. Also, execute the notebook.
- @05_eda_multivariate.ipynb Can you add a grouped box plot to the notebook showing Sales by Popularity level, with separate groups for Mobile and Tablet? I want to see whether the popularity pattern holds across both product categories. Also, execute the notebook.

1.7 Connecting EDA to the Model

- @05_eda_multivariate.ipynb Based on everything we have seen in EDA, including the correlation bar chart, the scatter plots, and the heatmap, can you give me a summary of the top three variables to use for Simple Linear Regression and the reasoning behind each choice? Also list which variables should go into the Multiple Linear Regression and flag anything worth being cautious about.

1.8 Train/Test Split and Encoding

- Create a new notebook named 06_encoding.ipynb and place it under the notebooks folder.
- @06_encoding.ipynb Can you split the processed dataset into a training set and a test set using an 80/20 split with random_state=42? Add the code to the notebook and print the number of rows in each set. The test set should be set aside from here. Also, save the datasets in the appropriate folder.
- @06_encoding.ipynb Now that the data has been split, we need to prepare the training and test sets for modelling. Can you suggest an encoding plan for all the categorical columns in the dataset? For each column, tell me what method you would use and why. Don't run anything yet — I want to review the plan first.

- @06_encoding.ipynb The plan looks good. Can you apply the encoding to both the training and test sets? Fit only on the training data and apply the same transformation to the test set. Save both encoded datasets to data/processed/ and print the final column list.

Note: Split must happen before encoding to avoid data leakage. Encoding is fitted on training data only and applied to the test set.

1.9 Simple Linear Regression

- Create a new notebook named 07_simple_lr.ipynb and place it under the notebooks folder.
- @07_simple_lr.ipynb Let's build the first simple linear regression model using statsmodels OLS, with Campaign_Engagement_Score as the predictor and Sales as the target. Fit it on the training data, add a scatter plot of the actual values with the regression line on top, and print the full model summary including the coefficient, intercept, p-value, R^2 , RMSE, MAE, and F-statistic.
- @07_simple_lr.ipynb Can you explain what the intercept value of \$8,046 means in this model, and whether it has a practical interpretation?
- What is the actual minimum value of Campaign_Engagement_Score in this dataset?
- @07_simple_lr.ipynb If a campaign's engagement score improves from 50 to 60, what does this model predict the change in Sales to be?
- What is the actual range of Campaign_Engagement_Score in this dataset? Confirm whether scores of 50 and 60 fall within the observed range.
- @07_simple_lr.ipynb Can you run the standard linear regression assumption checks and add the results to the notebook? For each one, explain what the result is telling us.
- @07_simple_lr.ipynb Same approach — simple linear regression with statsmodels OLS, Advertising_Expenditure as the predictor. Fit on training, add the scatter with the regression line, and print the full model summary.
- @07_simple_lr.ipynb The intercept here is $-\$7,464$. Can you explain what this means and whether it makes practical sense?
- What is the actual minimum value of Advertising_Expenditure in this dataset?
- @07_simple_lr.ipynb If ElectraHub increases the advertising budget for a campaign from \$600 to \$700, what does this model predict the impact on Sales to be? Also, what range of ad spend values can we trust this prediction within?
- Confirm whether \$600 and \$700 ad spend are within the observed range of Advertising_Expenditure in the training data.
- @07_simple_lr.ipynb Can you run the same assumption checks for this model?
- @07_simple_lr.ipynb One more simple model — Avg_Customer_Rating predicting Sales using statsmodels OLS. Fit on training, add the scatter with the regression line, and print the full model summary.
- What is the actual range of Avg_Customer_Rating in this dataset? What is the minimum and maximum?
- @07_simple_lr.ipynb The intercept here is $-\$69,698$. Can you explain what this means?
- What is the actual range of Avg_Customer_Rating in the training data? And if the slope is \$24,072 per star and the minimum rating is 3.57, at what rating would the line cross zero Sales?
- @07_simple_lr.ipynb If ElectraHub could improve the average customer rating of a product from 3.8 to 4.2, what does this model predict the change in Sales to be?
- Confirm whether ratings of 3.8 and 4.2 are within the observed range of Avg_Customer_Rating in the training data.
- @07_simple_lr.ipynb Same assumption checks for this model — Can you do it?
- @07_simple_lr.ipynb Can you add a summary table to the notebook comparing all three simple linear regression models? Show the predictor, coefficient, R^2 , RMSE, and MAE on the test set for each. Include a naive baseline row at the top for reference.

1.10 Multiple Linear Regression and Model Evaluation

- Create a new notebook named 08_multiple_lr.ipynb and place it under the notebooks folder.

- @08_multiple_lr.ipynb Let's build the full multiple linear regression model using statsmodels OLS with all the predictors in the encoded dataset. Fit it on the training data, evaluate on the test set, and print R^2 , Adjusted R^2 , RMSE, MAE, and F-statistic. Also print all coefficients with their p-values, sorted from largest to smallest by their absolute value.
- From the multivariate EDA we did earlier, what were the correlations of Inventory_Level, Num_Reviews, and Length_Product_Description with Sales?
- @08_multiple_lr.ipynb If a campaign's engagement score goes from 50 to 60, what does this model predict the change in Sales to be, holding everything else the same?
- @08_multiple_lr.ipynb What is the predicted Sales difference between running an Influencer campaign versus an Email campaign, with everything else equal?
- @08_multiple_lr.ipynb If a product moves from Low popularity to High popularity, what does the model predict the Sales increase to be?
- @08_multiple_lr.ipynb Can you run the assumption checks for this model and also produce a VIF table for all predictors?
- @08_multiple_lr.ipynb Can you add a final comparison table to the notebook showing all models side by side — the naive baseline, the three simple LR models, and the multiple LR — with R^2 , Adjusted R^2 , RMSE, and MAE on the test set?

1.11 Business Insights and Report Development

- Can you create a clean HTML report and save it to reports/insights_report.html? The report should include: a short summary of the business problem, the key EDA findings with the actual numbers, the model results with the full comparison table, three to four recommendations for the marketing and merchandising teams, and the model's known limitations. Keep the language simple and business-focused throughout — no technical jargon.